

- 485.** Total number of digits =
(No. of digits in 1-digit page nos. + No. of digits in 2-digit page nos. + No. of digits in 3-digit page nos.)
 $= (1 \times 9 + 2 \times 90 + 3 \times 267)$
 $= (9 + 180 + 801) = 990.$
- 486.** No. of digits in 1-digit page nos. $= 1 \times 9 = 9.$
 No. of digits in 2-digit page nos. $= 2 \times 90 = 180.$
 No. of digits in 3-digit page nos. $= 3 \times 900 = 2700.$
 No. of digits in 4-digit page nos. $= 3189 - (9 + 180 + 2700) = 3189 - 2889 = 300.$
 $\therefore$ No. of pages with 4-digit page nos. $= \left(\frac{300}{4}\right) = 75.$
 Hence, total number of pages $= (999 + 75) = 1074.$
- 487.** Let the number of students in sections A and B be x and y respectively. Then,
 $x + 10 = 3(y - 10) \Rightarrow 3y - x = 40 \quad \dots(i)$
 And, $x - 10 = y + 10 \Rightarrow x - y = 20 \quad \dots(ii)$
 Adding (i) and (ii), we get: $2y = 60$ or $y = 30.$
 Putting $y = 30$ in (ii), we get: $x = 50.$
- 488.** Clearly, the boy lost 7 chocolates, regained the 7 lost chocolates and won 20 more.
 $\therefore$ Number of games played $= (7 + 7 + 20) = 34.$
- 489.** Clearly, we have:
 $J - 40 = \frac{1}{2}A \quad \dots(i)$
 $A - 40 = J \quad \dots(ii)$
 $A - 40 = R + 40 \quad \dots(iii)$
 Solving (i) and (ii), we get: $J = 120$ and $A = 160.$
 Putting $A = 160$ in (iii), we get: $R = 80.$
 $\therefore$ Total money $= R + J + A = ₹(80 + 120 + 160) = ₹360.$
- 490.** We have:
 $B + 8 = C \quad \dots(i)$
 $A - 8 = C - 3 \quad \dots(ii)$
 $A + 6 = 2D \quad \dots(iii)$
 $B + D = 50 \quad \dots(iv)$
 Putting $C = A - 5$ from (ii) into (i), we have:
 $B + 8 = A - 5$ or $A - B = 13 \quad \dots(v)$
 Putting $D = 50 - B$ from (iv) into (iii), we have:
 $A + 6 = 100 - 2B$ or $A + 2B = 94 \quad \dots(vi)$
 Solving (v) and (vi), we get: $B = 27, A = 40.$
- 491.** We have:
 $B - 3 = E \quad \dots(i)$
 $B + 3 = D \quad \dots(ii)$
 $A + B = D + E + 10 \quad \dots(iii)$
 $B = C + 2 \quad \dots(iv)$
 $A + B + C + D + E = 133 \quad \dots(v)$
 From (i) and (ii), we have: $2B = D + E \quad \dots(vi)$
 Using (iii), (iv) and (vi) in (v), we get:
 $(B + 10) + B + (B - 2) + 2B = 133$
 $\Leftrightarrow 5B = 125$
 $\Leftrightarrow B = 25.$
- 492.** Since A and B each has ₹ 16 in the end, so C gave each of them ₹ 8. So, before third transaction, C had ₹ $(16 + 8 \times 2) = ₹ 32$ while A and B each had ₹ 8. Thus, in second transaction, B gave ₹ 4 to A and ₹ 16 to C. So, before second transaction, A had ₹ 4, B had ₹ $(8 + 4 + 16) = ₹ 28$ and C had ₹ 16. Thus, in first transaction, A gave ₹ 14 to B and ₹ 8 to C. So, before first transaction, A had ₹ $(4 + 14 + 8) = ₹ 26$, B had ₹ 14 and C had ₹ 8.

- 493.** Suppose Iqbal has x cards and Mushtaq has y cards.
 Let the number of cards given be $z.$
 Then, $x + z = 4(y - z) \Rightarrow x - 4y = -5z \quad \dots(i)$
 And, $x - z = 3(y + z) \Rightarrow x - 3y = 4z \quad \dots(ii)$
 Subtracting (i) from (ii), we get: $y = 9z.$ Putting $y = 9z$ in (i), we get: $x = 31z.$
 Clearly, since a pack of playing cards has 52 cards, z cannot be greater than 1.
 So, $x = 31$ and $y = 9 \quad [\because z = 1]$
 Hence, Iqbal had 31 cards.
- 494.** Let the original number of boys be $x.$ Then, number of apples first picked $= 3x.$
 Final number of boys $= x + 3.$ Final number of apples $= 3x + 1.$
 $\therefore \frac{3x + 1}{x + 3} = 2 \Leftrightarrow 3x + 1 = 2x + 6 \Leftrightarrow x = 5.$
 Hence, required number of apples $= 3 \times 5 + 1 = 16.$
- 495.** Let there be n students in the classroom.
 Total number of friendship bands bought $= n(n - 1).$
 If two students were absent, number of bands utilized $= (n - 2)(n - 3).$
 $\therefore n(n - 1) - (n - 2)(n - 3)$
 $= 122 \Leftrightarrow n^2 - n - n^2 + 5n - 6$
 $= 122 \Leftrightarrow 4n = 128 \Leftrightarrow n = 32.$
 So, number of bands finally unutilized $= 32 \times 31 - 31 \times 30 = 992 - 930 = 62.$
- 496.** Let the total number of chocolates she bought be x and the total number of her friends be $y.$
 Then, $3(y - 1) + 2 = x \Leftrightarrow x - 3y = -1 \quad \dots(i)$
 And, $2y + 15 = x \Leftrightarrow x - 2y = 15 \quad \dots(ii)$
 Subtracting (i) from (ii), we get: $y = 16.$ Putting $y = 16$ in (ii), we get: $x = 47.$
- 497.** Clearly, each committee has two members which are common to all three committees. In addition, it has a third member common to the second committee and a fourth one common to the third committee.
- 498.** $2n + 5m = 50 \Leftrightarrow 2n = 50 - 5m \Leftrightarrow n = \frac{50 - 5m}{2}.$
 Since n is a natural number, so $(50 - 5m)$ must be even and so, m must be even
 i.e., $m = 2, 4, 6$ or 8
 $[\because m < 10, \text{ as for } m = 10, n = 0]$
 If $m = 2, n = 20;$
 If $m = 4, n = 15;$
 If $m = 6, n = 10;$
 If $m = 8, n = 5.$
 $\therefore$ Least possible difference $= 8 - 5 = 3.$
- 499.** Let the number of chairs with arms bought be x and those without arms be $y.$
 Then, $x + y \geq 15$ or $y \geq 15 - x$
 And, $160x + 100y \leq 2000$
 $\Rightarrow 160x + 100(15 - x) \leq 2000$
 $[\because \text{For maximum value of } x, y \text{ has to be minimum}]$
 $\Rightarrow 160x + 1500 - 100x \leq 2000 \Rightarrow 60x \leq 500 \Rightarrow x \leq 8.$
 Hence, maximum value of x is 8.